# Deduce the World: How Far Can Reasoning Go?

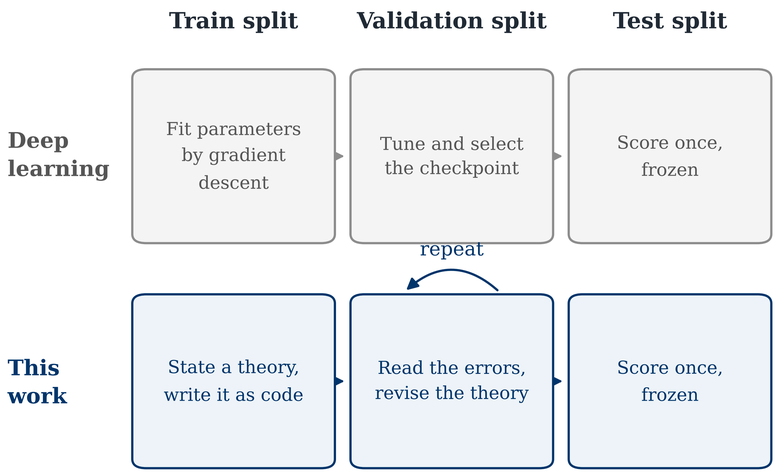
Jeffrey Wei — Yale University
Brian Cheung — University of California, San Francisco

Results & figures

## 1 Theories, not fitted predictors

Science deduces the world from first principles, revising its theories where their predictions fail.

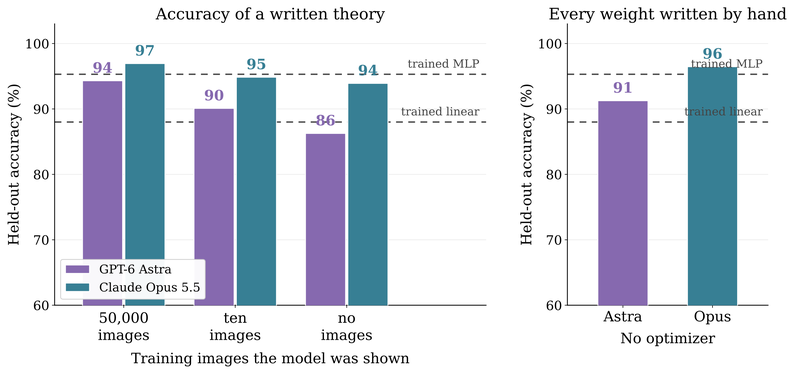
We ask whether a reasoning model can form scientific theories, and whether those theories generalize better than deep learning.




**May** read papers and the web, measure the training data, inspect its own errors

**May not** fit parameters to labels, search automatically over rules, store examples, see the test split

## 2 How much data a theory needs

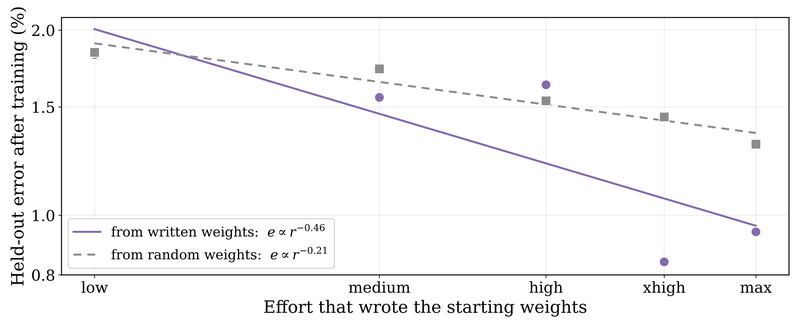



MNIST, 60,000 training and 10,000 test images. Best of five reasoning efforts; dashed lines are trained models.

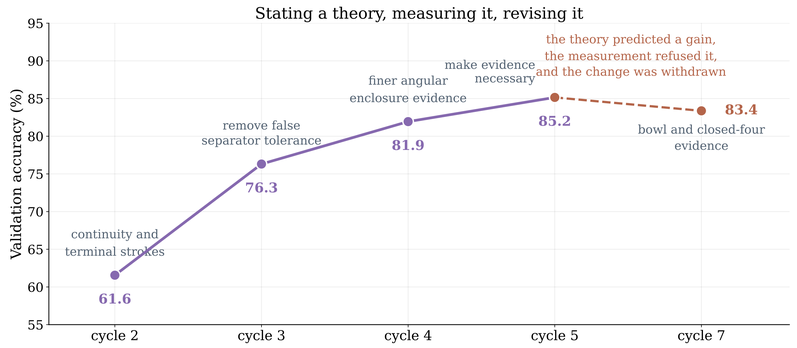
> Given no training images at all, a theory of how digits are written classifies most of them.

### Reasoning as initialization

Networks initialized from written weights train to lower error, and the gap widens with reasoning effort.



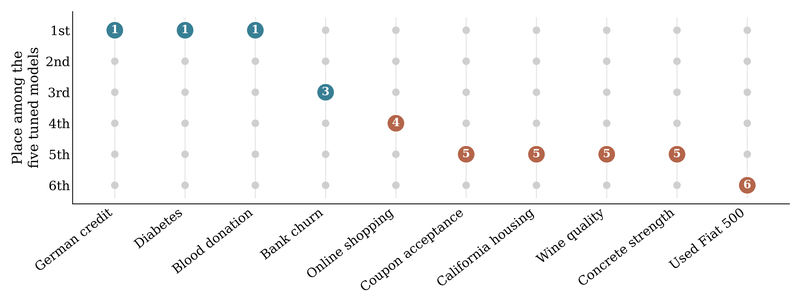

### The deduction–abduction loop




One digits session. Each label is the change the model argued for before measuring.

> It abduces a cause, deduces a change, and withdraws the change when the measurement refuses it.

## 3 The same protocol, ten fields

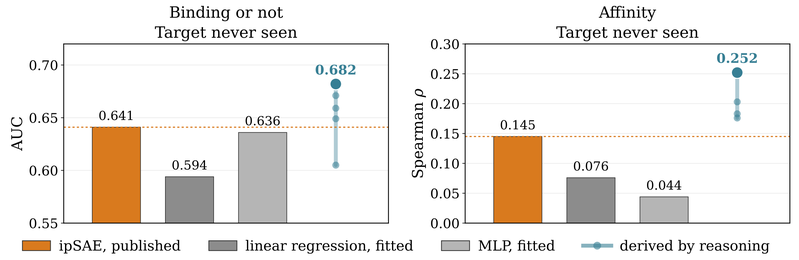



Ten tasks from TabArena, ranked against five tuned models on the same rows.

> Wins on small clinical tables, trails boosted trees on feature interactions.

## 4 Deriving a scientific score

Each session receives a structure predictor's error matrix and nothing else.




704 designed binders, holding out a whole target. Detection on the left, affinity on the right.

> A rule derived from physics beats the published score on a target it has never seen, and beats both fitted models.

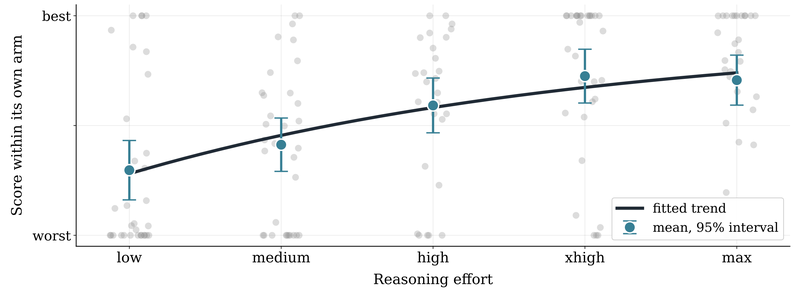
### The theory it stated

Treat the bound complex as a harmonic well. Equipartition makes the interface stiffness the inverse square of its fluctuation, and a stiffer well is a deeper one. Dividing by the binder's own internal error removes the predictor, and the units with it. Association confines six rigid-body degrees of freedom, each worth half the thermal energy times the log of the stiffness.

$$\kappa = \left(\frac{\sigma_0}{\sigma_{\text{iface}}}\right)^2, \qquad \mathrm{p}K_{\mathrm{d}} = \mathrm{p}K_{\mathrm{d,ref}} + 3\log_{10}\kappa.$$

The three is six halved. Nothing is fitted.

## 5 Scaling reasoning effort




## 6 So how far can reasoning go?

- Deep learning dominates the domains that are data-rich and theory-poor.
- Those are exactly the domains whose first principles nobody has written down yet.
- As reasoning improves, we expect models to supply them, and stated theories to replace fitted predictors there.